Sahaj Bhadja

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Summary

Software engineer with 2+ years of **software testing experience** including test automation frameworks, Python scripting, and CI/CD pipelines. Strong foundations in SDLC/STLC with proven ability to investigate defects, execute test suites, and drive quality improvements.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Data Structures, Software Engineering, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Object-Oriented Programming (C++), Data Structures, Software Testing, Algorithms

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Created and executed **test cases aligned with software requirements**, validating integration of software modules and verifying defect fixes prior to system-level deployment for production reliability.
- Continuously assessed **testing effectiveness** and recommended improvements by analyzing metrics throughout the SDLC, contributing to procedures and best practices that strengthened software quality.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Investigated, troubleshot, and resolved **software defects** in production systems handling 10K+ daily requests, analyzing results and documenting findings to support defect resolution and quality improvements.
- Developed **automated test tools** including Python scripts and test utilities, executing automated test suites and analyzing results to ensure robust and scalable testing across multiple features.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Developed **Python scripts and reusable code libraries** for data processing pipelines, applying software engineering quality processes to improve system performance by 18% through systematic analysis.
- Demonstrated strong **willingness to learn** new methodologies and tools, collaborating effectively with cross-functional teams while thriving in a results-driven research environment.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Designed and implemented **test automation frameworks** using Python and pytest, developing automated test tools and scripts that streamlined testing across multiple projects with 95% test coverage.
- Established **CI/CD pipelines using Jenkins**, integrating automated test execution and continuously assessing testing effectiveness through metrics analysis throughout SDLC/STLC phases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Created and maintained **test cases for software validation**, verifying defect fixes and performing integration testing prior to deployment while documenting findings for quality improvements.
- Built effective team collaboration with **strong communication and problem-solving skills**, contributing to refinement of testing procedures and standards that strengthened software reliability.

Skills

Languages: Python (scripting), C++ (academic), JavaScript, SQL
Testing: Test automation frameworks, unit testing (pytest, Jest), integration testing, test case design
Tools: Jenkins CI/CD, Git, Docker, VS Code, project management tools
Practices: SDLC/STLC, defect tracking, software quality processes, Agile/Scrum, documentation

Projects

Linguistix Code | *Python, Docker, Jenkins CI/CD, Automated Testing*
- Developed **test automation framework** with CI/CD pipelines, implementing automated test execution, validation checks, and deployment workflows to ensure software quality and reliability.

Big-Brain Live | *Python, PostgreSQL, Algorithm Design*
- Built software with comprehensive **test suites and validation procedures**, applying systematic testing approaches to achieve 95% accuracy across 50K+ document queries through iterative quality improvements.

Chat2SQL Code | *Python, MySQL, Test Automation*
- Implemented **software testing and validation** processes, creating test cases aligned with requirements and executing integration testing to verify module functionality before deployment.